

LESSON 2-9 • SESSION 2 • PREREQUISITE PRACTICE B • SKILL 1 OF 2

Find the MAD

Builds on Lesson 2-9 Session 1 • 6.SP.B.5c

TODAY'S LESSON

In Problem 2 you find the MAD of Trail A and the MAD of Trail B before you compare them.

YOU NEED TO

Find the mean absolute deviation (MAD): the mean of the distances from the mean. **You will use it in Practice Form A, problems 2 and 5.**

STEPS TO FOLLOW*use these same steps on every problem*

- 1 Find** each value's distance from the mean.
- 2 Add** the distances.
- 3 Divide** by the number of values.

WATCH ME*the steps in action***The mean of 66, 70, 74, 78 is 72. Find the MAD.****Step 1** Distances: 6, 2, 2, 6.**Step 2** $6 + 2 + 2 + 6 = 16$.**Step 3** $16 \div 4 = 4$.**Answer 4****TRY IT WITH HELP***fill in each step*

1. The mean of 29, 33, 37, 41 is 35. Find the MAD.

Step 1 Distances: _____**Step 2** $6 + 2 + 2 + 6 =$ _____**Step 3** $16 \div 4 =$ _____

2. The mean of 6, 8, 10, 12 is 9. Find the MAD.

Step 1 Distances: _____**Step 2** $3 + 1 + 1 + 3 =$ _____**Step 3** $8 \div 4 =$ _____**YOUR TURN***use the steps on your own*

3. Find the MAD. The mean is given.

a. 7, 8, 9, 8 (mean 8) _____**b.** 4, 10, 6, 12 (mean 8) _____**c.** 40, 46, 54, 60 (mean 50) _____**d.** 43, 43, 57, 57 (mean 50) _____

Explain What a MAD Means

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TODAY'S LESSON

Team N's MAD of 3 means its scores sit about 3 points from 82.

YOU NEED TOExplain a MAD: the typical distance from the mean. **You will use it in Practice Form A, problems 4 and 7.****STEPS TO FOLLOW***use these same steps on every problem*

- 1 Name** the mean and the MAD.
- 2 Say:** the values are typically about MAD away from the mean.
- 3 Compare:** a smaller MAD means steadier values.

WATCH ME*the steps in action***Team N has mean 82 and MAD 3. What does the MAD mean?**

- Step 1** Mean 82, MAD 3.
- Step 2** Scores are typically about 3 points from 82.
- Step 3** That is close, so Team N is steady.

Answer Scores are typically about 3 points from 82.**TRY IT WITH HELP***fill in each step*

- 4.** Times have mean 45 min and MAD 3 min. What does the MAD mean?

- Step 1** Mean _____, MAD _____.
- Step 2** Times are typically about _____ min from 45.
- Step 3** Is 3 min close or far? _____

- 5.** Set A has MAD 2. Set B has MAD 9. Which is steadier?

- Step 1** MADs: _____ and _____.
- Step 2** A's values are typically about _____ from its mean.
- Step 3** The smaller MAD is steadier: Set _____.

YOUR TURN*use the steps on your own*

- 6.** Which set is steadier?

- a.** MAD 14 or MAD 3 _____
- b.** MAD 0.5 or MAD 3 _____
- c.** MAD 4 or MAD 9 _____
- d.** MAD 6 or MAD 11 _____

READY CHECK*a problem like the one in today's lesson*

- 7.** Supplier 1 has MAD 0.5 day. Supplier 2 has MAD 3 days. Which is true?

- A** Supplier 1 varies less.
- B** Supplier 2 varies less.
- C** They vary the same.
- D** You cannot tell.

CHECK YOURSELF

- ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____